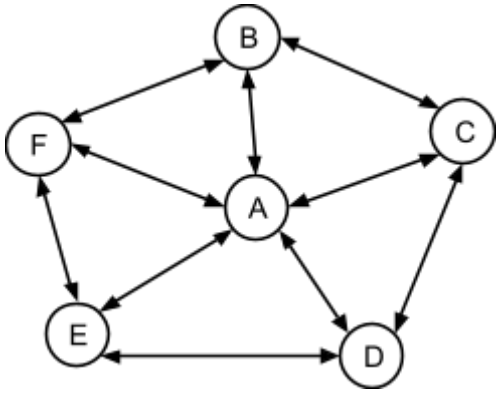
# American Computer Science League

2024-2025 ● Contest 4: Short Problems ● Junior Division

**1. Graph Theory**

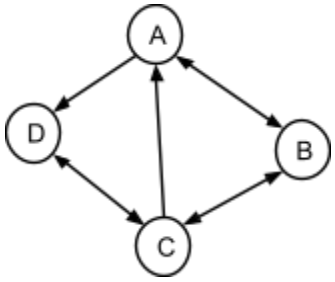
How many 1s are in the adjacency matrix for the following graph?




A. 10
B. 15
C. 20
D. 30
E. 36

**2. Graph Theory**

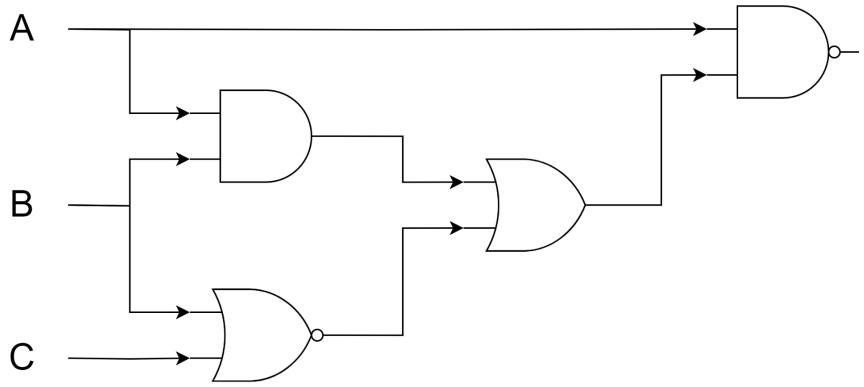
How many cycles are there in the directed graph below?




A. 3
B. 4
C. 5
D. 6
E. 7

**3. Digital Electronics**

Translate the following digital circuit into a Boolean expression.




A. $\overline{A\,(AB\,+\overline{\overline{B}\,+\,C})}$

B. $\overline{A\,(A\,+\,B\,+\overline{BC})}$

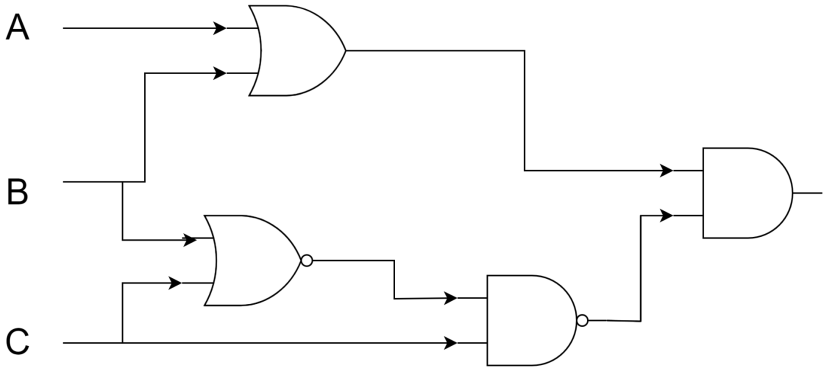
C. $\overline{\overline{A}\,(AB\,+\,\overline{\overline{B}\,+\,C})}$

D. $\overline{A\,+\,AB\,+\overline{BC}}$

E. $\overline{A\,(\overline{AB}\,+\overline{B\,+\,C})}$

**4. Digital Electronics**

How many ordered triples make the following digital circuit TRUE?




A. 2
B. 3
C. 4
D. 5
E. 6

**5. What Does This Program Do? (Strings)**

What will be the output for the following program?

```
s = "JavaPythonC++"
ans1 = s[3:]
ans2 = s[4:9]
ans3 = s[:4]
output ans2 + ans1 + ans3
```

A. `Pytho++JavaP`
B. `PythoC++Jav`
C. `PythonJavaC++`
D. `C++PythonJava`
E. `PythonC++Java`

**6. What Does This Program Do? (Strings)**

What will be the output for the following program?

```
s = "summeriscoming"
ans = ""
for x = 0 to len(s)/2 - 1
     y = len(s) - x - 1
     if s[x] < s[y] then
          ans = ans + s[x]
     else
          ans = ans + s[y]
     end if
next x
output ans
```

A. `gnimeci`
B. `gnimec`
C. `summors`
D. `gnieci`
E. `gnimocsiremmus`